Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

DEPARTAMENTO DE FÍSICA

TRABAJO FIN DE GRADO

Sensibilidades a radio de carga de neutrinos en experimentos de neutrinos de reactor involucrando CEνNS

Realizado por
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PHYSICS DEPARTMENT

Sensitivities to neutrino charge radius at reactor neutrino experiments involving CEνNS

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agradecimiento aqui.

Resumen

resumen aqui

Abstract

resumen en ingles

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1. Introduction

intro

2. Electroweak theory in the Standard Model of Particles

2.1. Introduction

The Standard Model of particles (SM) is the most currently utilized theory of interactions of subatomic particles. It describes all matter and forces between them, excluding gravity, as interactions between 12 fermions (6 quarks and 6 leptons) with their corresponding anti-fermions, 4 spin-1 force-carrying gauge bosons and 1 scalar boson as shown in Fig. 2.1. Within this framework exist 3 distinct forces: strong nuclear force, mediated by the gluon (g) boson; electromagnetic (EM) force, mediated by the photon (γ) boson; and weak nuclear force, mediated by the $W^{\pm}$ and Z bosons. This work pertains to the description of the properties of the only family of particles that present no electric or color charge known as neutrinos (ν), wherein the SM exist three known flavors (ν_e, ν_μ and ν_τ) corresponding to the latter half of (electro)weak isospin doublets formed with the known charged leptons (e^-, μ^-, τ^-). This chapter aims to describe neutrinos in the SM, the weak force and its origin as a part of a larger broken symmetry known as "Electroweak Theory" (EW).

2.2. Symmetries and fermion content

2.2.1 Neutrinos

2.2.2 Weak Nuclear Force

2.3. Massive and Massless Mediators

2.4. Electroweak mixing $\sin^2\theta_W$

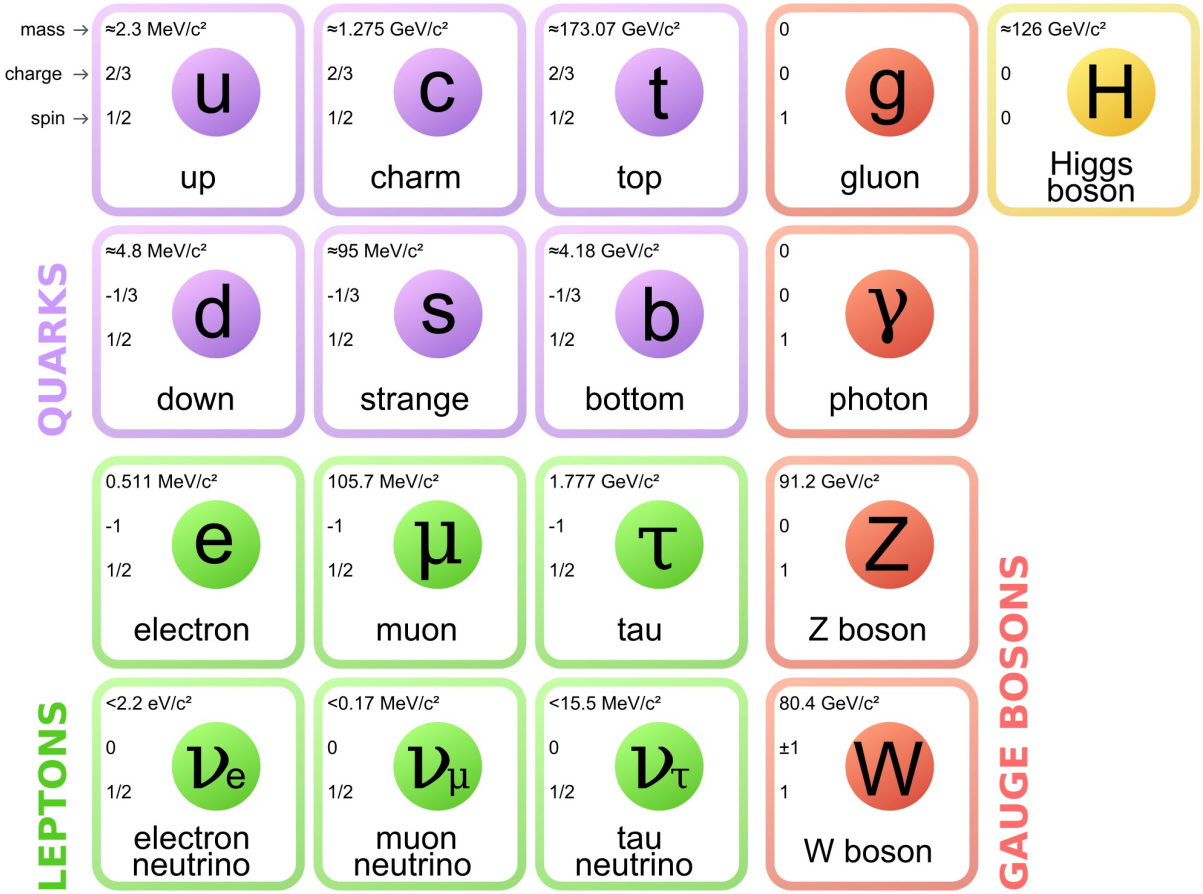

Figure 2.1: *Particles in the SM*

3. Radiative Corrections and Neutrino Charge Radius

3.1. Introduction

3.2. Neutrino Charge Radius

3.3. One-loop Corrections to the $\nu\nu\gamma$ vertex

3.4. NCR as a radiative correction of Electroweak Mixing

4. Coherent Elastic Neutrino-Nucleus Scattering

4.1. Introduction

4.2. Cross Section

4.3. Experiments

4.4. Quenching Factors

4.5. Neutrino Sources

4.6. Form Factors

5. SBC-ININ-LAr

5.1. Results

6. Summary and conclusions